Lecture 9 2025-09-25

Topics for midterm 1

① Calculus Review (Ch 1.1.)

- Continuity, MVT/Rolle's theorem, EVT, IVT, integrals
- Taylor's theorem

② Floating point numbers (Ch 1.2)

- IEEE standard
- Conversion binary ↔ decimal
- Rounding
- Cancellation of digits
- Relative and absolute errors

③ Algorithms and convergence (Ch 1.3)

- What is an algorithm, pseudocode?
- Stability and error propagation
- Rates of convergence

④ Solving nonlinear equations ($f(x)=0$, $g(x)=x$)
(Ch 2.1, 2.2)

- bisection method
- fixed point iteration

Back to fixed point iteration

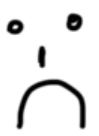
Is $|g'(x)| < 1$ crucial?

Let's try $g(x) = 2x$ $\quad (g(x) = 2x = x)$

unique fixed point: $p = 0$

If we start with $p_0 \neq 0$, then the iterates grow rapidly:

$$p_0 = 1$$
$$p_1 = g(p_0) = 2p_0 = 2$$
$$p_2 = g(p_1) = 2p_1 = 4$$
$$p_3 = g(p_2) = 2 \cdot p_2 = 8$$
$$\vdots$$



Can we fix this? We are interested in a fixed point of $2x = x$

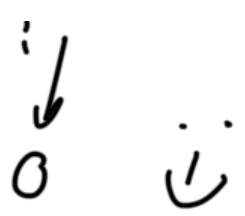
which we can write as $x = \frac{x}{2}$

Define $\tilde{g}(x) = \frac{x}{2}$

Then fixed points of g are fixed points of $\tilde{g}$ and vice versa

$$|\tilde{g}'(x)| = \frac{1}{2} < 1$$

$$p_0 = 1$$
$$p_1 = \tilde{g}(p_0) = \frac{1}{2}$$
$$p_2 = \tilde{g}(p_1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$
$$p_3 = \tilde{g}(p_2) = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$$
$$\vdots$$



In Matlab implementation, we used stopping criterion from Cor.3 which is a good estimator unless k is close to 1

$$|p_n - p| \leq \frac{k}{1-k} \underbrace{|p_n - p_{n-1}|}_{\leq \text{tol (in Matlab)}}$$

More examples in textbook and on homework.

2.3 Newton's Method

We go back to solving $f(x) = 0$

We assume $f \in C^2([a,b])$ and $p_0 \in [a,b]$ approximates the solution p of $f(x)=0$ with $f'(p_0) \neq 0$ ($|p - p_0|$ "small")

We use Taylor's theorem:

$$f(p) = f(p_0) + (p-p_0) f'(p_0) + \frac{(p-p_0)^2}{2} f''(\xi(p))$$

We set $f(p)=0$ and $(p-p_0)^2$ is negligible (if $|p-p_0|<1$ small)

$$0 \approx f(p_0) + (p-p_0) f'(p_0) \quad \text{which we can solve for } p$$

$$p \approx p_1 = p_0 - \frac{f(p_0)}{f'(p_0)}$$

So we can define a sequence $\{p_n\}_{n=1}^{\infty}$ by

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}$$

$f'(p_{n-1})$

This is Newton's method.

Input: Initial guess p_0, tolerance ε, maximum number of iterations N_{max}

Output: Approximate p of $f(p)=0$ or message failure.

Step 1: Set $i=1$

Step 2: while $i \le N_{max}$ do

- Set $p = p_0 - \frac{f(p_0)}{f'(p_0)}$ (compute p_i)
- If $|p-p_0| < \varepsilon$ then
  return p;
- Set $i = i+1$
- Set $p_0 = p$ (update p_0)

Step 3: return "The method failed to converge after N_{max} iterations"

Stopping criteria:

* $|p_n - p_{n-1}| < \varepsilon$ (what we used in Matlab)
* $\frac{|p_n - p_{n-1}|}{|p_n|} < \varepsilon$
* $|f(p_n)| < \varepsilon$

Theorem 2.6 (Convergence of Newton's Method)

Let $f \in C^2([a,b])$, $p \in (a,b)$ with $f(p)=0$, $f'(p) \neq 0$

Then $\exists \delta > 0$ such that $p_n \to p$ provided $p_0 \in [p-\delta, p+\delta]$.

("If you start close enough, Newton's method converges")

Proof: Idea: Recast as a fixed point iteration $p_{n+1} = g(p_n)$

with $g(x) = x - \dfrac{f(x)}{f'(x)}$

(Note $g(x) = x \iff f(x) = 0$)

We have $g'(x) = 1 - \dfrac{(f'(x))^2 - f(x)f''(x)}{(f'(x))^2} = \dfrac{f(x)f''(x)}{(f'(x))^2}$

If $f(p) = 0 \Rightarrow g'(p) = 0$ (o)